



VLN-CE

"Walk past the couch ..."



ObjectNav

"Find the chair"



Embodied QA

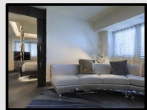
"What color is the picture?"



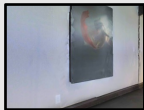
Aerial-VLN

"The target is a sedan ..."

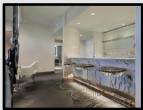
Current Observation



Front



Left



Right



Back

Navigation History



Language Action

Tool Call



turn(direction)



backtrack(waypoint)



go_stair(up/down)



double_check(stop)

online
update



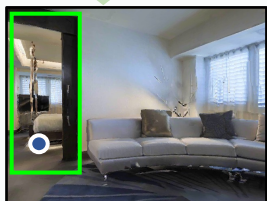
Vision Action



select(bbox/point)



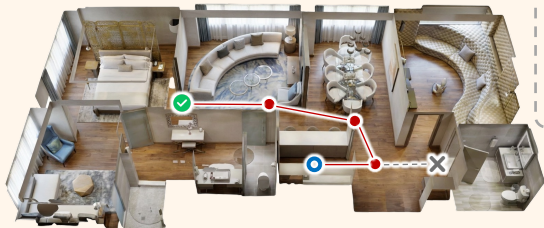
target_desc(text)



doorway to the bedroom



Robot Action



- Start
- Failed Path
- Failed
- Success Path
- Goal

new
waypoint



TODO List Memory



Go straight and turn left



Walk between the pillar and the island



Head towards the bedroom



Wait in front of the tables by the couch

add()

remove()

rewrite()

check()

if backtrack needed



Second-Chance Backtrack

Backtracks to the last confident waypoint to have a second chance

Failed route



Last confident
waypoint



New success route

